RV INSTITUTE OF TECHNOLOGY AND MANAGEMENT®

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DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

PROJECT SYNOPSIS

on

“RESEARCHMIND X”

Agentic AI Research Intelligence and Research Gap Discovery Platform

Submitted in partial fulfilment of the requirements for the Mini/Major Project
Master of Computer Applications (MCA)

Submitted by

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1. Title of the Project

“ResearchMind X: Agentic AI Research Intelligence and Research Gap Discovery Platform” — an end-to-end AI research intelligence system that accepts research topics, questions, keywords, and academic documents; analyzes scholarly literature using NLP, semantic retrieval, RAG and agentic workflows; compares methodologies, datasets, findings and limitations; and generates evidence-supported candidate research gaps and research questions.

2. Introduction

The rapid growth of scientific publications has made literature review increasingly difficult for students and researchers. Finding papers is only the first step: researchers must read multiple documents, extract important information, compare methodologies and datasets, understand conflicting findings, identify limitations, and determine whether an area remains insufficiently explored. These activities are often repetitive and time-consuming when performed manually.

ResearchMind X is proposed as an AI-powered research intelligence platform that brings these activities into a unified workflow. A user can begin with a research topic or question and optionally provide papers, URLs/DOIs, keywords, publication constraints, or an existing collection of documents. The system processes the literature, creates a searchable semantic knowledge base, retrieves relevant evidence, performs cross-paper analysis, and generates a structured research intelligence report. Its gap-detection component produces candidate research gaps based on observed evidence and recurring patterns; it does not claim to automatically prove scientific novelty.

3. Problem Statement

Students and researchers face difficulty in efficiently analyzing large collections of academic literature, extracting comparable information from papers, identifying recurring methodological limitations, and discovering potentially underexplored research areas. Existing academic search tools mainly retrieve publications, while conventional summarization or PDF-chat systems often focus on individual documents and do not provide systematic cross-paper analysis. Therefore, there is a need for an integrated AI-based system that can discover relevant literature, understand research documents, organize evidence, compare studies, identify patterns and limitations, and generate traceable candidate research gaps and research questions for human evaluation.

4. Motivation

- The increasing volume of research literature makes manual literature discovery and comparison time-consuming.
- Students need a systematic way to understand what has already been studied before selecting a project or dissertation problem.
- Important research information is distributed across abstracts, methodologies, datasets, results, limitations, conclusions, and references.
- Semantic retrieval can identify conceptually related evidence even when the exact keywords differ.
- RAG can ground generated responses in retrieved literature and provide source traceability.
- Agentic workflows can divide a complex research task into literature discovery, document analysis, evidence verification, gap analysis, and report generation.
- The project provides a strong demonstration of modern AIML and AI-engineering skills including NLP, embeddings, vector databases, RAG, LLMs, agents, evaluation, APIs, databases and deployment.

5. Objectives

- To accept research topics, research questions, keywords, paper files, and supported paper identifiers as inputs.
- To collect, clean, parse, and organize academic documents with useful bibliographic and research metadata.
- To extract research entities such as methodology, datasets, evaluation metrics, findings, limitations, and future-work statements.
- To generate semantic embeddings and create a searchable research knowledge base using vector retrieval.

- To implement a retrieval-augmented generation pipeline that produces evidence-grounded responses with source references.
- To compare papers across methodologies, datasets, results, evaluation measures, limitations, and research directions.
- To detect recurring limitations, underrepresented approaches, inconsistent findings, and other evidence-supported patterns that may indicate candidate research gaps.
- To generate potential research questions and a structured research intelligence report for researcher review.
- To evaluate the system using retrieval relevance, groundedness, citation correctness, response quality, latency, and other suitable metrics.

6. Literature Survey

Modern scholarly information systems increasingly combine large-scale academic metadata, semantic retrieval, citation networks, natural language processing, and generative AI. Scholarly APIs and knowledge graphs such as OpenAlex provide structured information about works, authors, institutions, topics, and citation relationships, supporting large-scale literature discovery and analysis. Semantic Scholar provides programmatic access to scholarly paper and citation information and can support literature exploration.

Transformer-based language models and sentence-embedding approaches have improved semantic understanding of text, allowing systems to retrieve conceptually related passages instead of depending only on exact keyword matching. Retrieval-Augmented Generation combines external evidence retrieval with language generation and is useful for knowledge-intensive question answering. However, generated research insights must remain traceable to their evidence and should be reviewed by a human researcher. ResearchMind X therefore focuses on combining literature retrieval, document intelligence, cross-paper comparison, evidence tracking, and candidate-gap discovery in a single research workflow.

7. Proposed System

ResearchMind X begins with a user research request. The request may be a topic, research question, keywords, uploaded PDFs, paper URL/DOI, or a combination of these. The system discovers or imports relevant literature, extracts document text and metadata, cleans and segments the content, and generates embeddings for semantic retrieval. The processed information is stored together with bibliographic metadata and source locations.

For an analysis request, the system retrieves the most relevant evidence and supplies it to the RAG pipeline. Cross-paper analysis components organize comparable information such as methodology, datasets, results, metrics and limitations. An agentic workflow coordinates literature discovery, paper analysis, evidence verification, candidate-gap generation and report creation. The final output provides summaries, comparisons, recurring limitations, candidate research gaps, supporting evidence, potential research questions and references.

8. System Architecture / Modules

8.1 Research Input & Literature Acquisition Module

Accepts research topics, research questions, keywords, uploaded PDFs, paper URLs/DOIs, and optional constraints such as publication period or domain. It imports user documents and/or retrieves suitable literature from supported scholarly sources while maintaining bibliographic metadata.

8.2 Document Processing & Research Information Extraction Module

Extracts text and document structure from academic papers and identifies useful sections such as abstract, introduction, methodology, datasets, results, discussion, limitations, conclusion and references. The module cleans and normalizes the content for downstream processing.

8.3 Semantic Embedding & Research Knowledge Base Module

Splits documents into meaningful chunks and converts them into embedding vectors. Embeddings, document identifiers, page/section information and metadata are stored in a vector-searchable knowledge base for semantic retrieval.

8.4 Retrieval-Augmented Generation Module

Retrieves relevant evidence for each research query and supplies the evidence as context to the language model. The response is connected to source documents so the researcher can inspect the supporting material.

8.5 Cross-Paper Comparison Module

Converts extracted information into comparable research attributes and presents structured comparisons of methods, datasets, evaluation metrics, findings, limitations and future work. This module supports both tabular and visual literature analysis.

8.6 Research Gap Detection & Evidence Analysis Module

Analyzes recurring limitations, underrepresented methods or datasets, inconsistent findings, missing evaluation conditions, and other patterns across the selected literature. Candidate gaps are generated together with supporting papers or evidence passages so that users can review them.

8.7 Agentic Research Workflow Module

Coordinates specialized workflow stages such as Literature Discovery Agent, Paper Analysis Agent, Evidence Verification Agent, Gap Analysis Agent and Report Generation Agent. Each stage produces structured intermediate results for the next stage.

8.8 Research Intelligence Dashboard

Provides an interactive interface for uploading papers, entering research questions, viewing literature summaries, comparing studies, exploring evidence, reviewing candidate gaps, and generating a final research intelligence report.

9. Technology Stack

- Programming Language: Python
- Frontend: React.js, TypeScript, Tailwind CSS
- Backend: FastAPI
- Database: PostgreSQL
- Vector Search: FAISS / Chroma or another suitable vector database
- NLP / Embeddings: Sentence Transformers, Hugging Face models
- Generative AI: suitable LLM API or local LLM
- RAG / Agent Orchestration: LangChain/LangGraph or equivalent workflow framework
- Document Processing: PyMuPDF and suitable PDF/text extraction libraries
- Data & Evaluation: Pandas, NumPy, scikit-learn
- Visualization: Matplotlib / Plotly / Recharts
- Scholarly Sources: OpenAlex, Semantic Scholar and other permitted academic sources
- Development & Deployment: Git, GitHub, Docker and a suitable cloud deployment platform

10. Expected Outcomes

- A functional research intelligence platform that accepts multiple forms of research input including topics, questions, keywords and academic documents.
- A searchable knowledge base containing processed research text, metadata, embeddings and source references.
- Semantic literature retrieval and RAG-based research answers grounded in retrieved evidence.
- Automatic structured summaries and cross-paper comparison of methodologies, datasets, results, metrics and limitations.
- Evidence-supported identification of recurring limitations and candidate research gaps across a selected literature collection.
- Generation of potential research questions and a structured research intelligence report for human review.
- A measurable evaluation framework for retrieval quality, groundedness, citation correctness, response quality and system performance.

- A deployable, portfolio-ready AIML application demonstrating NLP, information retrieval, embeddings, RAG, LLMs, agentic AI, backend APIs, databases and visualization.

11. Applications

- Literature review and research-topic exploration for MCA, engineering, postgraduate and undergraduate students.
- Research proposal and dissertation preparation support.
- Research supervision and literature comparison support for faculty members.
- Academic research landscape analysis across a selected domain.
- Technology and R&D; literature monitoring for organizations.
- Identification of recurring research limitations and candidate research directions within a defined literature collection.

12. Advantages

- Reduces repetitive manual effort involved in organizing and comparing research papers.
- Provides semantic literature discovery rather than relying only on exact keyword matches.
- Combines information from multiple papers into structured cross-paper analysis.
- Maintains evidence and source traceability for important generated findings.
- Separates candidate research-gap generation from the final human judgment of scientific novelty.
- Modular architecture allows independent improvement of models, retrieval, database, agents and interface.
- Demonstrates a broad range of current AIML and AI-engineering skills in one coherent project.

13. Limitations & Future Scope

System performance depends on the quality, accessibility, coverage and diversity of the available literature. A candidate research gap generated by the system cannot guarantee scientific novelty because novelty assessment requires comprehensive literature coverage and expert judgment. Poorly structured PDFs, OCR errors, inaccessible full texts, incomplete metadata, ambiguous citations and domain-specific terminology may affect extraction and retrieval quality. LLM-generated content can also contain errors; therefore, important conclusions should remain linked to source evidence and be reviewed by a researcher.

Future scope includes multimodal understanding of research figures and tables, citation-network visualization, temporal research-trend analysis, systematic-review assistance, multilingual literature processing, domain-specific research agents, improved scientific-information extraction, private/local LLM deployment, stronger automated evaluation benchmarks, institutional repository integration, and optional collaborative research workspaces.

14. References

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